

# Knowledge Graph-Informed Query Decomposition(KG-IQD): Hybrid KG-RAG Reasoning in Noisy Contexts

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## Abstract

Disaster data is inherently noisy and heterogeneous, posing significant challenges for automated reasoning. Retrieval-Augmented Generation (RAG) systems retrieve relevant documents but often fail to capture relational structure, impairing tasks such as comparative analysis and summarization, critical for disaster response. Knowledge graphs (KGs), while structured, are often sparse and lack contextual depth, which may not allow them to capture in-depth information needed for reasoning and planning queries. We propose Knowledge Graph-Informed Query Decomposition (KG-IQD), a hybrid KG-RAG framework that uses KGs to strategically guide the query decomposition into a set of focused, single-answer, contextually relevant subqueries, which perform the highly targeted RAG process, ensuring that the final LLM-generated answer is coherently grounded in both the retrieved textual evidence and the rich structured context of the KG. We evaluated KG-IQD against strong baselines using our novel Indian Disaster Dataset and an "LLM-as-judge" framework. To ensure a rigorous and fair evaluation, our methodology mitigates positional bias through output randomization and employs a metric that penalizes verbosity and irrelevance. The results demonstrate a clear advantage: KG-IQD outperforms RQ-RAG by 14% and KG by 18%. Our dataset, code, and output are publicly available at: <https://github.com/hexronuspi/kg-iqd>.

## Keywords

Knowledge Graph, Disaster Data, Information Retrieval, Query Optimization

## 1. Introduction

The information about an ongoing disaster event or any similar historical events is essential for effective disaster response and planning. Such information is available and distributed over the web in various formats, such as government reports, news feeds, images, etc. Effectively synthesizing such distributed information is important, yet challenging due to the inherent heterogeneity in the data. While Large Language Models (LLMs) have emerged as powerful tools for processing text (potentially unstructured), their direct application often yields verbose, generic answers while missing out the specific details, such as "For how many days a specific area was flooded during a past flood event". The more structured approaches, such as Retrieval-Augmented Generation (RAG) and Knowledge Graphs [1] (KGs) have demonstrated better question-answering performance by providing relevant information about a specific topic in a structured form. However, our preliminary analysis shows that these methods (i.e., RAG and KG) also possess fundamental limitations.

The standard RAG architectures [2] treat documents as unstructured text and retrieve relevant passages from a corpus but lack structural awareness for complex queries. They often fail to model entity relationships, limiting performance on tasks like comparative analysis or multi-faceted summarization[3]. For instance, RAG can retrieve facts about two separate disaster events but struggles with comparative queries. In contrast, KGs explicitly represent atomic facts about entities and their relationships (e.g.,  $\langle \text{Nepal\_Earthquake\_2015}, \text{Economic\_Loss}, 10\text{B USD} \rangle$ ). KG construction extracts such facts from source texts but discards their contextual richness[4]. In summary, RAG offers better contextual depth for single events but struggles with synthesis across events, while KGs support comparison but may lack context.

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To appear

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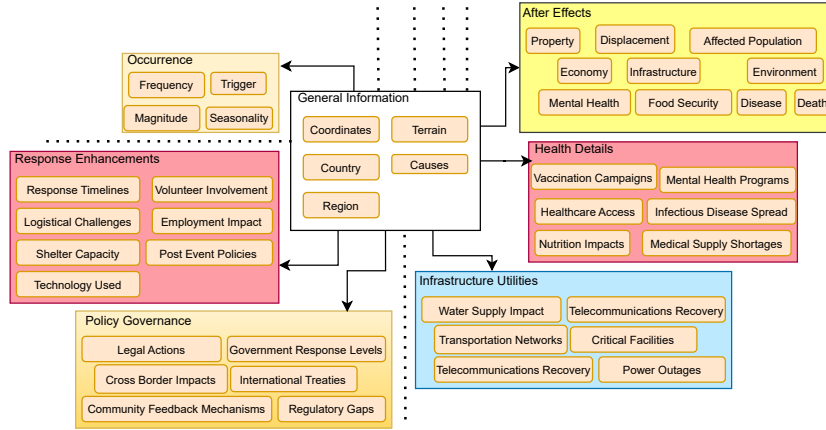


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To bridge this critical gap, we introduce Knowledge Graph-Informed Query Decomposition (KG-IQD), a novel hybrid framework that harmonizes the information from KGs with the contextual depth of RAG system. Specifically, our approach uses the KG to strategically guide the RAG in the context retrieval process. The core of our contribution is a query decomposition [5] technique that is inspired by RQ-RAG [6]. KG-IQD retrieves relevant triples from the KG and uses them to decompose a complex query into focused, single-answer sub-queries. These guide a targeted RAG process over the text corpus, ensuring that retrieved chunks align with the user’s structural intent. This hybrid approach overcomes the limitations of using RAG or KG alone, enabling precise reasoning for comparative and summary-based disaster queries where existing RQ-RAG and KG methods fall short.

## 2. Dataset

We curated specialized corpora of historical disaster events from publicly available, reputable sources, including government assessments, NGO reports, and academic studies. For the work presented in this paper, the corpus collected contains details of 2015 Nepal Earthquake and 2018 Kerala Flood. We developed a KG, a hierarchical schema to model the complex entities and relationships to describe disaster events. This schema comprises **43 classes** (e.g., ‘DisasterEvent’, ‘Infrastructure’, ‘ResponseEffort’) and **207 relationships** that capture fine-grained details such as causality, location, and quantitative impact. We employed LLM (Gemini 2.5 Pro) to perform a zero-shot, schema-guided extraction of facts (triples), populating our KG from the unstructured text corpora. Figure 1 shows a snippet of the conceptual schema used for KG generation. We used Gemini 2.5 Pro due to the sheer token length (>1M) it provides, which allowed us to pass our entire data set for a single disaster in one go and populate the entire (43, 207) KG. The LLM output (i.e. individual triples) was verified through human evaluation to make sure the generated KG is accurate, exhibiting no hallucinations or deviations from the original text corpora.



**Figure 1:** Knowledge Graph Schema Illustration

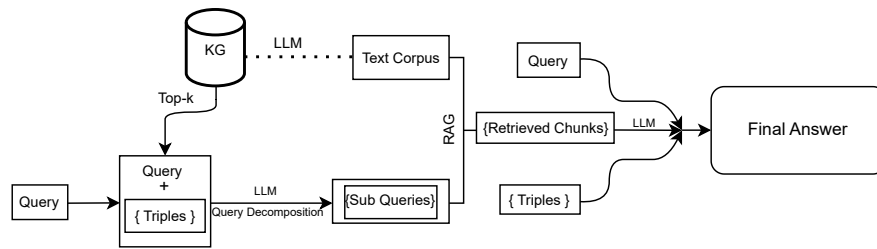
## 3. Methodology

The objective of the proposed approach is to guide the RAG system in the retrieval process by efficiently decomposing (or rewriting) a complex query into a set of a few simple, targeted, and relevant queries. When employing LLM for query decomposition or rewriting, in an unguided scenario, there is a risk of generating semantically similar but less relevant queries. For example, consider the query “What were the challenges faced by people after Nepal Earthquake?”. For this query, the after-effects describing “the impacts on the tourism economy” and “blocked roads due to landslides” are more relevant to the Nepal earthquake than “IT infrastructure failure” that may be relevant to the disasters in San Francisco,

USA. Conventional unguided decomposition techniques [5] are typically domain-agnostic and may generate irrelevant or incoherent sub-queries. The inclusion of less relevant queries may dilute the LLM’s context window, which will lead to incomplete answers. Our key contribution, KG-guided query decomposition, addresses this limitation by leveraging the facts from KG to rewrite precise sub-queries. The proposed method constrains the query decomposition to factually relevant and domain-specific elements, preserving both semantic alignment and grounding fidelity, which is critical for accurate generation. This approach ensures that RAG outputs are not only context-rich but also sharply focused and highly reliable.

### 3.1. KG-IQD Framework

Figure 2 shows the proposed KG-IQD framework that treats the retrieval as a multi-stage process. Instead of a monolithic semantic search, it uses a KG to guide and constrain the retrieval process, ensuring both precision and depth in the generated answers.



**Figure 2:** KG-IQD Architecture. The query is decomposed into sub-queries using top-k KG triples passed through an LLM. Retrieved text chunks and triples are fused with the query to generate the final answer.

**KG-Guided Query Decomposition:** KG-IQD first grounds the user query by retrieving relevant triples from our disaster-specific KG. This structured context provides a semantically rich prior that guides an LLM in decomposing the initial complex prompt into a series of focused, single-answer sub-queries. Each sub-query is tightly aligned with the factual basis encoded in the KG, reducing noise and ambiguity in downstream retrieval.

**Targeted Retrieval and Synthesis:** These sub-queries then drive a targeted RAG process that retrieves high-precision evidence from the raw text corpus. The final answer is synthesized by an LLM that integrates information from both the retrieved textual context and the original KG grounding (for summary tasks), enabling robust, reasoning grounded in structured and unstructured knowledge alike.

## 3.2. Experiments

### 3.2.1. Evaluation Tasks

We designed evaluation queries ranked by reasoning complexity. Medium-complexity queries test contextual and comparative reasoning, while high-complexity queries probe causal inference, planning, and multi-hop reasoning capabilities.

**Table 1**

Evaluation Queries by Reasoning Complexity. ‘A’: 2015 Nepal Earthquake, ‘B’: 2018 Kerala Floods

| Category                                                | Query                                                                                         |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Medium Complexity (Contextual &amp; Comparative)</b> |                                                                                               |
| M1                                                      | Which event had a greater impact on healthcare access: ‘A’ or the ‘B’?                        |
| M2                                                      | Compare the environmental and socio-economic impacts between the ‘A’ and the ‘B’.             |
| <b>High Complexity (Causal, Planning, Multi-hop)</b>    |                                                                                               |
| H1                                                      | Which mitigation measures and community engagement strategies were most effective during ‘A’? |
| H2                                                      | What logistical and infrastructure challenges affected evacuation efforts during ‘B’?         |

3.2.2. Baselines

We evaluate our method on complex reasoning tasks, excluding trivial fact-based queries(eg: What was the magnitude of the Nepal earthquake in 2015?) and baseline RAG as it fails on tasks such as summarization and comparative analysis. Three systems—KG-only, RQ-RAG, and base LLM—are benchmarked on four curated disaster-specific queries spanning two difficulty levels: (1) Contextual & Comparative (Medium), and (2) Multi-hop, Planning & Causal (High). Each query targets a model’s ability to synthesize insights from structured and unstructured sources, beyond basic retrieval.

3.2.3. Evaluation Methodology: Penalty-Aware LLM Judge

To ensure objective and reproducible evaluation, we adapted the LLM-as-Judge [7] framework by including a penalty mechanism. The judge is governed by a strict, zero-shot rubric designed to reward data-driven intelligence. Its core directive is to maximize a score based on high factual density (e.g., statistics, named entities) while penalizing generic, verbose, and unsubstantiated claims. To mitigate any positional bias by the judge LLM during evaluation, the order of the four model outputs (KG, RAG, LLM(Gemini 2.5 Pro), KG-IQD) was randomized for each adjudication trial.

Table 2  
Evaluation Criteria and Weights

| Criterion                     | Weight | Description                                          |
|-------------------------------|--------|------------------------------------------------------|
| Factual Density & Specificity | 60%    | Rewards specifics; penalizes generic, vague content. |
| Efficiency & Clarity          | 20%    | Rewards concise, unambiguous, and fluent responses.  |
| Relevance & Topical Alignment | 20%    | How directly the response addresses the user query.  |

4. Results and Analysis

The results in Table 3 show KG-IQD consistently outperforms all baselines, achieving perfect scores on both query types. The base LLM performs worst, reflecting its generic, ungrounded outputs. The human validation confirmed KG-IQD’s advantage. While these results are promising, in the future, we will further evaluate the performance and robustness with the corpus of a large number of heterogeneous disaster events.

Table 3  
LLM-as-Judge scores across all queries. KG-IQD consistently outperforms all baselines, achieving a perfect score. Best scores are in bold.

| Query ID        | KG-Only | RQ-RAG | Base LLM | KG-IQD(Ours) |
|-----------------|---------|--------|----------|--------------|
| M1 (Healthcare) | 9.3     | 7.5    | 4.5      | <b>10.0</b>  |
| M2 (Impacts)    | 8.0     | 9.0    | 6.0      | <b>10.0</b>  |
| H1 (Mitigation) | 9.4     | 9.7    | 3.8      | <b>10.0</b>  |
| H2 (Logistics)  | 7.0     | 9.0    | 5.0      | <b>10.0</b>  |

5. Limitations & Future Work.

While effective on small-scale graphs, our current approach is untested on web-scale data and relies on manual schema design. Future work will address these limitations by developing scalable, distributed systems for real-time graph construction and retrieval.

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